

Application of AI in Modern Chatbot Systems

Abstract

This paper discusses the application of artificial intelligence in chatbot systems, focusing on retrieval-augmented generation and large language models.

Introduction

Chatbots have evolved significantly with the introduction of AI techniques such as natural language processing and deep learning.

Methodology

We use a RAG pipeline combining document loaders, embeddings, and vector databases to retrieve relevant context.

Results

The system shows improved accuracy in answering domain-specific queries compared to traditional chatbots.

Conclusion

AI-powered chatbots provide scalable and efficient solutions for information retrieval and user interaction.